In Situ Defect Annealing of Swift Heavy Ion Irradiated CeO_2 and ThO_2 Using Synchrotron X-Ray Diffraction

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Abstract

Hydrothermal diamond anvil cells (HDACs) provide facile means for coupling synchrotron X-ray techniques with pressure up to 10 GPa and temperature up to 1300 K. This manuscript reports on an application of the HDAC as an ambient-pressure sample environment for performing *in situ* defect annealing and thermal expansion studies of swift heavy ion irradiated CeO_2 and ThO_2 using synchrotron X-ray diffraction. The advantages of the *in situ* HDAC technique over conventional annealing methods include: rapid temperature ramping and quench times, high-resolution measurement capability, simultaneous annealing of multiple samples, and prolonged temperature- and apparatus stability at high temperatures. Isochronal annealing between 300 K and 1100 K revealed 2-stage and 1-stage defect recovery processes for irradiated CeO_2 and ThO_2, respectively; indicating that the morphology of the defects produced by swift heavy ion irradiation of these two materials differs significantly. These results suggest that electronic configuration plays a major role in both the radiation-induced defect production and high temperature defect recovery mechanisms of CeO_2 and ThO_2.

Introduction

Diamond-anvil cells (DACs) provide unparalleled high-pressure capabilities and are indispensible tools in diverse scientific fields including chemistry [1], geology [2], and materials science [3]. Of further utility is the combination of DACs with *in situ* application of high temperature and high-energy radiation. Coupling such extreme conditions can induce highly nonequilibrium, transient states that open pathways to novel phase transformations and modifications in materials [4]. Modern techniques for combining high pressure and high temperature environments involve the use of DACs with laser- and electrical heating systems.

Laser-heating systems can reach temperatures in excess of 5000 K and maintain minimal temperature fluctuations [5, 6]. These features, in conjunction with *in situ* synchrotron X-ray measurement capabilities, make laser-heated DAC systems ideal for probing the physics of extreme environments. However, laser-based systems require extensive optical alignment and calibration to yield stable heating and measurement conditions, and temperature measurements are less reliable below 1500 K. Electrical heating systems are complimentary to the laser heating method because they can generate and maintain temperatures from 300 K – 1200 K for several hours [7]. A unique electrical heating apparatus, the hydrothermal diamond-anvil cell (HDAC), offers great flexibility with respect to sample type and experimental conditions.

Hydrothermal DACs incorporate resistive heating elements and thermocouples within the pressure apparatus, thereby eliminating the additional complexity associated with external alignment and calibration. In addition, the HDAC design minimizes heat loss through superior thermal insulation of the sample chamber such that 150 W is sufficient to reach sample temperatures of ~1300 K [8]. In the HDAC configuration, the wire-wrapped diamond-anvils are the source of heating. The two anvils are individually heated and put in close contact (~100 μm) with each other to surround the sample chamber and yield stable, homogenous heating conditions. The simplicity of operation of the HDAC and the ease of access to intermediate temperature regimes (300 - 1300 K) makes this an attractive tool for performing *in situ* thermal annealing studies using synchrotron X-ray sources. Such annealing experiments are valuable for investigating defect morphology, diffusion kinetics, and unit cell recovery, among other applications [9, 10]. Thus, annealing experiments are vital analysis techniques for the development of current and next-generation energy materials. These materials require resiliency under high-energy irradiation, high temperature, and/or high-stress environments; hence, it is

important to characterize the evolution and recovery of novel materials *in situ* under these conditions. In this study, the combined use of an HDAC and synchrotron X-ray diffraction (XRD) was applied to swift heavy ion-irradiated CeO_2 and ThO_2 for nuclear fuels investigations.

Cerium dioxide (CeO_2) has a fluorite-structure, which is the same as that of typical nuclear fuels such as UO_2 [11]. Thorium dioxide is a proposed LWR fuel material that boasts superior thermo-physical properties, chemical stability, and proliferation resistance properties over traditional UO_2 and PuO_2 fuels [12]. During reactor operation, nuclear fuels are exposed to intense radiation fields and temperature gradients. These conditions, in particular fission-fragment- and fast neutron-fluxes, generate nanoscale defects and induce adverse effects such as swelling and redox reactions [13, 14]. Such modifications can degrade materials properties relevant to fuel performance [15]. Much work has been dedicated towards investigating the behavior of nuclear fuel materials under neutron and α-particle irradiations; however, studies concerning the effect of high-energy fission-fragment bombardment in fuel materials are far more limited.

Swift heavy ions have been used to simulate the effect of ionizing fission-fragment irradiation because of the similarity in specific kinetic energies (>1 MeV/nucleon) and energy deposition mechanisms. Swift heavy ions deposit substantial quantities of energy in the electronic system of the target in extremely short (sub-picosecond) timescales triggering electron excitation and ionization events, followed by phonon emission during decay of electrons from the conduction band [16]. Depending on the material, these inelastic interaction processes can result in complex damage morphologies. In CeO_2, for example, heavy-ion irradiation induces anion deficiency and cation valence reduction in cylindrical damage zones (called "ion tracks") while maintaining bulk integrity of the fluorite-structure [11, 14, 17]. In ThO_2, spectroscopic

studies indicate that cation chemistry is unaffected and charge modifications in the local cation environment are balanced by long-range ionic interactions [18]. These results elucidate the response of actinide- and structurally related oxides to swift heavy ion irradiation, yet key questions remain unanswered, such as: What are the kinetics of damage recovery in swift heavy-ion irradiated oxides? Furthermore, does the defect morphology and do defect migration energies in ThO_2, with non-multivalent Th^{4+} cations, differ from CeO_2, UO_2, and PuO_2, which possess multivalent cations?

This paper introduces an HDAC-synchrotron XRD experimental setup that enables *in situ* investigations of the structure and recovery kinetics in a vast array of materials at ambient pressure. Specific examples demonstrate the precision of this technique and the potential applications to energy materials investigations. First, the thermal expansion of swift heavy ion-irradiated ThO_2 was measured *in situ* at high temperature. Second, the defect recovery kinetics of swift heavy ion-irradiated CeO_2 and ThO_2 were characterized *via* simultaneous, *in situ* isochronal annealing of both materials.

Experimental

Polycrystalline pellets of CeO_2 and ThO_2, with typical grain sizes on the order of 1 μm, were prepared by pressing powders into 200 μm diameter holes that were drilled into 50 μm thick stainless steel foils. The samples were irradiated at room temperature and under vacuum at the M2 beamline of the UNILAC at the GSI Helmholtzzentrum für Schwerionenforschung in Darmstadt, Germany with 945 MeV ^{197}Au ions to a fluence of 2.5×10^{13} cm^{-2}. The energy loss and range of the ions in both materials was determined using the SRIM 2013 code [19] assuming 60 percent theoretical density [20]. Density-corrected SRIM calculations [21] estimate that 945

MeV ^{197}Au ions traverse though the entirety of the pressed pellets and exhibit a mean electronic energy loss per ion of 24 keV/nm in both CeO_2 and ThO_2. The uncertainty in these calculated values is approximately 10 percent. The nuclear energy loss along the entire ion path is approximately three orders of magnitude lower than the electronic energy loss and was therefore considered negligible. Additional details regarding the sample preparation and irradiation techniques are described elsewhere [22].

After irradiation, the sample compacts were extracted from the stainless-steel foil holders using a needle. From each sample, a micro-grain was transferred into one of two holes (50-100 μm diameter) that were drilled into a rhenium gasket (Fig. 1) for use with a Basset-modified HDAC [8]. The gasket was pre-indented to a thickness of ~100 μm using the two diamond-anvils with 400 μm cutlet size. The two holes had different diameters to facilitate the identification of the two samples in the chamber during the synchrotron x-ray measurements. The loaded gasket was sealed between the diamond-anvils without exerting pressure on the samples.

Angle-dispersive XRD was performed at the High Pressure Collaborative Access Team (HPCAT) 16-BM-D beam line at the Advanced Photon Source at Argonne National Laboratory. A monochromatic 25 keV (λ = 0.4959 Å) beam selected by a Si (111) double crystal monochromator with a focused spot size of 12 μm (v) × 5 μm (h) was used in transmission geometry. Debye-Scherrer rings were measured using a Mar345 image plate detector. Two-dimensional X-ray transmission scans were performed at the beam line to verify the integrity of the samples in the HDAC and to align the X-ray beam with the samples (Fig. 1). The HDAC was connected to an Ar-1%H_2 gas source to cool and prevent oxidation of the molybdenum heating coils and the diamond-anvils (Fig. 2). The metal HDAC holder was cooled with circulating water. The temperature of each diamond-anvil was controlled both individually and remotely

outside the beamline hutch using one of two Zantrex XHR 33-33 (0 – 33 V; 0 – 33 A) power supplies, and the temperature of each diamond-anvil was monitored with thermocouples located in close proximity (~50 microns) to the tip of each diamond-anvil. The high thermal conductivity of the diamond-anvils resulted in minimal temperature variance between the thermocouple contact points and the culet adjacent to the sample, allowing for accurate tracking of the sample annealing temperature.

To determine the thermal expansion of irradiated ThO_2, diffraction patterns were measured *in situ* (i.e., at high temperature) at 50 K intervals using a collection time of 180 seconds as the temperature of the samples was continuously increased from 300 K to 875 K. Each temperature was maintained for 17 minutes prior to collecting a pattern and ramping to the next temperature step at a rate of 0.5 ºC/sec. To monitor the recovery kinetics of ion-beam induced defects in CeO_2 and ThO_2, the samples were isochronally annealed for 20 minutes in 50 - 75 K temperature steps from 300 K to 1075 K. After each heating step, the samples were quenched to ambient temperature and left to cool for an additional 20 minutes. Upon reaching stable temperature, diffraction patterns were recorded using a collection time of 300 seconds. Diffraction patterns were converted into integrated diffractograms using the software Fit2D [23] and unit cell parameter values were determined by Rietveld refinement [24].

Figure 3 shows the time required to reach and quench back to ambient from 875 K. Due to the high thermal conductivity of the diamonds, the average temperature ramp and quench times were 1-2 minutes. One advantage of such rapid ramp and quench times is that defect annealing can be studied under well-controlled conditions, which enables higher precision in the determination of activation energies. The stabilization of high temperatures in the HDAC was nearly instantaneous and temperature fluctuations were minimal. The average uncertainty in

annealing temperature (i.e., the temperature difference between the diamonds) during the experiments was ± 3°C. Visual inspection of the diamond-anvils before and after the experiments revealed that they exhibited exceptional stability during operation, even under prolonged use at high temperatures. The maximum temperature reached in the present study was 1075 K.

Results and Discussion

Thermal Expansion

Upon continuous heating, the unit cell of irradiated ThO_2 expands with increasing temperature (Fig. 4). The red data point represents the unit cell parameter after quenching from the highest temperature. Error bars derived from Rietveld refinement of the diffractograms are smaller than the data points plotted and are therefore not visible. The unit cell parameter of the unirradiated reference sample was determined to be 5.5970 ± 0.0001 Å (not shown here), which is in good agreement with the value of 5.5974 ± 0.0006 Å reported by Yamashita *et* al. [25]. The unit cell parameter after quenching, 5.6014 ± 0.0001 Å, was larger than the unirradiated value, indicating that radiation-induced defects were not fully annealed upon reaching 875 K —the maximum temperature in the thermal expansion study. A third-order polynomial was fit to the thermal expansion data to obtain a relation for the unit cell parameter as a function of absolute temperature (dashed line in Fig. 4). The fitted model of thermal expansion of irradiated ThO_2 (present study) agrees with the thermal expansion behavior of unirradiated ThO_2 measured by Yamashita *et al.* using a different technique, to within experimental uncertainty [25].

It is important to note the superior measurement sensitivity of *in situ* synchrotron XRD measurements, which is manifested in the root mean square of the error (RMSE) from the data

fitting. The RMSE of the present data (1.2×10^{-4}) is two orders of magnitude smaller than the estimated standard deviation (ESD) reported by Yamashita *et* al. (6.3×10^{-2}). This improved resolution is attributed to use of synchrotron XRD in combination with the optimum temperature conditions provided by the heatable diamond-anvil cell. This enables equations of state and thermal-expansion coefficients to be determined with a higher precision, allowing for the detection of miniscule changes in the thermoelastic properties of radiation-damaged materials. The HDAC method is particularly well suited to characterize changes in the thermoelastic properties of neutron-irradiated nuclear materials (for which the damage mechanisms are different and the damage often of greater magnitude than that of swift heavy ion irradiations) because the size of activated (radioactive) samples can be limited to microgram quantities without sacrificing data quality.

Isochronal Defect Annealing

Heavy ion irradiation induces an increase in unit cell parameter (i.e., unit cell expansion) due to the accumulation of both isolated point defects and agglomerated defect clusters. As a material is heated, these defects are annealed and the unit cell parameter begins to recover. To investigate the defect recovery kinetics in CeO_2 and ThO_2, changes in unit cell parameter were monitored as a function of isochronal annealing temperature (Fig. 5). The dashed trend lines in Fig. 5 are approximate and are used to guide the eye. The lines were plotted with slope values less than or equal to zero on the assumption that thermal expansion effects (unit cell expansion) were negligible in comparison to defect annealing effects (unit cell reduction) because of the rapid temperature ramp and quench times. Both materials exhibit a monotonic decrease in unit cell parameter as a function of annealing temperature, although the reduction in unit cell

parameters occurs at different rates. For CeO_2 (blue diamonds), unit cell parameter recovery begins at very low temperatures whereas, for ThO_2 (green squares), the unit cell parameter is approximately constant up to ~650 K, where recovery begins.

Williamson-Hall plots were constructed from the diffractograms collected at various annealing temperatures to determine if the unit cell parameter reduction was influenced by changes in heterogeneous microstrain in the materials (Fig. 6). Williamson-Hall analysis evaluates the dependence of peak broadening on the diffraction angle, θ, to decouple the individual contributions of grain size and microstrain to peak broadening [26]. Plotting $\beta cos(\theta)$ against $sin(\theta)$, where β is the radiation-induced peak broadening (the increase in full width at half-maximum), yields a line whose slope is proportional to the $tan(\theta)$ dependence of microstrain-induced peak broadening and whose y-intercept is inversely proportional to the $cos^{-1}(\theta)$ dependent crystallite size-induced peak broadening. The analysis revealed a negligible change in the average crystallite size (y-intercepts) as a function of annealing temperature in both CeO_2 and ThO_2. This demonstrates that the reduction of unit cell parameters at low temperatures is accompanied by the relaxation of heterogeneous microstrain in the material but that, at these temperatures, grain growth of the irradiated materials is negligible. Figure 6 also illustrates a large discrepancy in the irradiation-induced microstrain in the materials. The irradiation-induced microstrain in CeO_2 is a factor of two greater than that of ThO_2, with an augmented contribution to the microstrain arising from the anion sublattice, as alluded by the consistent offset of oxygen-contribution data points from the trend lines (see Fig. 6 caption).

Recovery stages and their corresponding activation energies were investigated using the derivative method [9, 27]. By this method, distinct defect annealing processes are identified *via* peaks in the differential curve of the data. The differential curves demonstrate that there is only

one defect recovery stage in ThO_2 while there exist two in CeO_2 in the temperature regime investigated (see Fig. 5 inset). The deviation implies a more complex defect annealing behavior in CeO_2, which is consistent with the Williamson-Hall analysis and the reported difference in irradiation-induced redox response of the two materials. X-ray absorption spectroscopy (XAS) and X-ray photoelectron spectroscopy (XPS) measurements of CeO_2 demonstrate that, under swift heavy ion irradiation, cations in the ion-track core partially reduce from Ce^{4+} to Ce^{3+} to offset local charge imbalance induced by anion-interstitial displacement to the periphery of the ion-track [17, 28, 29]. A similar redox response is notably absent in swift heavy ion irradiated ThO_2 due to the non-multivalent state of Th^{4+} cations [18]. Thus, the discrepancy in the magnitude of microstrain and the additional recovery region in the differential curve of CeO_2 can be related to (*i*) partial or full re-oxidation of Ce^{3+} atoms to the Ce^{4+} state and/or (*ii*) the annealing of defect structures that are not formed in ThO_2.

Re-oxidation is possible considering that the annealing was performed in an air environment, i.e., the samples were loaded into the gasket in air without a medium. A change from Ce^{3+} to Ce^{4+} would be accompanied by a decrease in ionic radius (1.14 Å to 0.97 Å) [30], which would further accelerate reduction of the average unit cell parameter. The elimination of trivalent cerium from the material, the presence of which causes a size mismatch on the cation sublattice and gives rise to local structural distortion, would also correlate to a reduction in the local strain field, which might manifest as a deviation(s) in the microstrain *vs.* temperature curve. However, combined influences on the strain field reduction by ionic radii changes and agglomerate defect annealing make it difficult to decouple individual contributions without XAS measurements and/or the use of an alternate sample environment.

Molecular dynamics simulations and theoretical calculations of fluorite structures show that the formation of cation defects is energetically unfavorable and the probability of defect mixing, in which a cation occupies an anion site (or *vice versa*), is low [31-33]. Hence, defects in CeO_2 and ThO_2 are predominantly anti-Frenkel defects (anion interstitial-vacancy pairs) with minor populations of Frenkel (cation pairs) and hole defects. The cylindrical ion-track structure in the oxides is similar to the core-shell morphology found in ionic alkali halide crystals [34, 35] and fluorite-derivative pyrochlores [36]. In CeO_2, the two regions exhibit differences in stoichiometry induced by the formation of oxygen vacancy-rich cores and oxygen interstitial-rich shells [11, 28]. In ThO_2, the concentration and size of defects decreases radially from the ion-track core [18]. In both materials, however, computational studies suggest the formation of charge-neutral oxygen defect clusters including molecular oxygen, split interstitials, and dislocation loops [37].

To elucidate the defect morphology of the irradiated samples, activation energies were derived from the differential curves and compared to literature values for specific defect types. The temperatures corresponding to the peaks in the differential recovery curve (see inset of Fig. 5) were taken to be average temperature values representative of the entire, respective recovery regions (regions *a*, *b*, and *c* in Fig. 5). These temperatures (380 K and 820 K for CeO_2, and 750 K for ThO_2) were used in the expression [38]:

$$E = kTln(Ct) \qquad (1)$$

where k is the Boltzmann constant and C is a frequency factor, to estimate the activation energy, E, for defect recovery occurring at a particular annealing temperature T and time t. If the frequency factor is known, Eq. (1) can be used to determine the most probable activation energy corresponding to the temperature for maximum recovery in each annealing stage under

isochronal annealing. In this study, t = 20 minutes and it was assumed that $C = 10^{10}$ s^{-1}, as suggested by Weber for the isostructural oxides [10]. The derived activation energies are 0.99 eV, 2.13 eV, and 1.95 eV for recovery regions *a*, *b*, and *c*, respectively (see Fig. 5).

The values for CeO_2, 0.99 eV and 2.13 eV, are consistent with the values calculated by Weber for O-interstitial migration and Ce-vacancy migration mechanisms in polycrystalline samples [10]. The value for ThO_2, 1.95 eV, does not agree with reported values for migration energies [31]; however, the differential peak of ThO_2 is much broader than the peaks of CeO_2, suggesting that there exists a distribution of energies, rather than one unique activation energy, for the recovery stage. Assuming the existence of multiple energies for the ThO_2 recovery stage, the recovery of ThO_2 can be attributed to both O-vacancy and O-interstitial migration. Cation interstitial and vacancy migration are not likely because the stable valence of Th^{4+} makes anion disorder predominate and cation interstitials seemingly play no part in the defect structure of ThO_2 [31, 39].

The derived activation energies are inconsistent with density functional theory (DFT) calculations by Xiao *et al*. that predict higher intrinsic mobility of oxygen defects in ThO_2 as opposed to CeO_2 [37]; however, discrepancies in migration energies may be attributed to irradiation and annealing conditions. Zinkle *et al*. indicate that ionizing radiation can promote the recovery of displacement damage in ceramic insulators by enhancing the mobility of point defects, in particular vacancies, *via* ionization-induced diffusion [40]. This suggests that the behavior of defects produced by highly ionizing radiation can differ from the behavior of intrinsic defects or defects that would result from other kinds of radiation, e.g., from ballistic processes. The charge states of oxygen interstitials and thus the energy of the migration barrier are also highly dependent on the sample environment. For example, Erhart and Albe found that

when annealing ZnO in an oxygen-rich atmosphere, oxygen diffusion was dominated by O^{2-} interstitials, as opposed to the O^0 and O^{2+} species, which resulted in a lower oxygen migration barrier [41]. Lastly, it should be noted that the redox behavior exhibited by CeO_2 under energetic irradiation conditions greatly influences defect behavior [14]. The complex defect partitioning and subsequent changes in ionic radii of cations in ion tracks yields concentration gradients that might in turn modify defect migration energies. A recent STEM study of swift heavy ion irradiated CeO_2 shows that in addition to O displacement, comparatively minor populations of Ce are displaced from the ion track core [28]. In ThO_2, there is no direct evidence of significant Th displacement. Studies suggest that the coordination of Th remains stable under irradiation and/or local changes in coordination are balanced by long-range ionic interactions [18]. These studies, in conjunction with studies citing relatively high cation defect formation and migration energies [31] might explain why cation-vacancy migration is observed in CeO_2 but not ThO_2

In summary, swift heavy ion irradiated CeO_2 and ThO_2 exhibit different defect recovery behaviors. CeO_2 exhibits a more complex annealing behavior than ThO_2, as indicated by the larger degree of heterogenous microstrain and the additional recovery stage observed during isochronal annealing. Unit cell volume recovery of irradiated CeO_2 and ThO_2 during the isochronal annealing can be attributed to the relaxation of heterogeneous microstrain and the recombination and annihilation of anion defects (regions *a* and *c* in Fig. 5). This is followed by cation vacancy migration in CeO_2 (region *b* in Fig. 5). Additional high-temperature *in situ* XAS [42, 43] and/or different annealing atmospheres are needed in order to confirm whether the recovery behavior in CeO_2 (region *b* in Fig. 5) is significantly modified by the partial/full re-oxidation of Ce^{3+} cations that are produced by ionizing radiation.

Conclusions

The combined use of an HDAC apparatus with synchrotron XRD was utilized to perform *in situ* isochronal annealing and thermal expansion studies on swift heavy ion irradiated CeO_2 and ThO_2. The advantages of the *in situ* HDAC technique over conventional annealing methods include: rapid temperature ramping and quench times, high-resolution measurement capability, simultaneous annealing of multiple samples, and prolonged temperature- and apparatus stability at high temperatures. The features, considering the potential for coupling with X-ray absorption spectroscopy and high-pressure, make the HDAC an attractive tool for performing high-resolution, *in situ* investigations of defect annealing, thermal expansion, and diffusion for a wide range of materials. Isochronal defect annealing in the HDAC independently confirmed the recently reported difference in defect behavior between swift heavy ion irradiated CeO_2 and ThO_2. CeO_2 exhibits a 2-stage defect recovery mechanism whereas ThO_2 exhibits a 1-stage mechanism. These results suggest that cation electronic configuration plays a significant role in not only the defect production behavior, but also the defect recovery mechanisms of the fluorite-structure oxides.

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Figure 1: Optical microscope (left) and X-ray transmission scan (right) images of the hydrothermal diamond-anvil cell sample chambers. Irradiated CeO_2 and ThO_2, were simultaneously annealed under identical sample conditions using a single rhenium gasket with two holes that served as sample chambers. The two-dimensional X-ray transmission scan was used to locate the samples in the gasket for aligning the X-ray beam with the samples. The X-ray absorption of the sample chamber is lower than that from the surrounding rhenium gasket as indicated by the transmitted X-ray intensity (red: high intensity *vs.* blue: low intensity).

Figure 2: The experimental hydrothermal diamond anvil cell (HDAC) setup at the HPCAT 16BM-D beam line of the Advanced Photon Source at Argonne National Laboratory. The upper left frame shows an identical setup with a Mar165 CCD detector. The HDAC is positioned in the synchrotron x-ray beam, such that the single crystal diamonds serve as windows into the sample containment cell. The upper right frame shows *in situ* heating of the sample *via* resistive heating of the diamond-anvils. The bottom frame schematically illustrates the HDAC assembly and the x-ray diffraction setup.

Figure 3: Temperature readings from one of the diamond-anvil thermocouples during the heating and cooling cycles. The red line (overlaid on the blue line for comparison) shows the ramping of temperature to a constant value prior to annealing. The blue line shows the quenching of sample temperature after prolonged annealing at a constant, elevated temperature. The high thermal conductivity of the diamond-anvils yields average heating and quench times of 1-2 minutes and minimal temperature fluctuation.

Figure 4: The thermal expansion of swift heavy-ion irradiated ThO_2 measured in a hydrothermal diamond-anvil cell (blue squares) and the associated third-order polynomial fit to the data (dashed black line). The coefficients of the polynomial fit are shown in the top left of the figure. The red point denotes the unit cell parameter after quenching from 875 K. The integrated diffractograms for the highest temperature measurement (875 K) and for the quenched sample measurement are compared at the bottom right of the figure.

Figure 5: The relative change in unit cell parameter of CeO_2 (blue diamonds) and ThO_2 (green squares) as a function of annealing temperature. The dashed trend lines are approximate and are used to guide the eye. The lines were plotted with slope values less than or equal to zero on the assumption that thermal expansion effects (unit cell expansion) were negligible in comparison to defect annealing effects (unit cell reduction) for the procedure (sequential heating and quenching) and ramp/quench times used. The inset shows the differential curves of the annealing data. The temperature values corresponding to the maxima of the derivative curves were used as representative values for estimating the activation energies of the three recovery regions (peaks). The linear recovery regions labeled *a*, *b*, and *c* correspond to the peaks labeled *a*, *b*, and *c* in the differential curves.

Figure 6: Williamson-Hall plots of irradiated CeO_2 (left) and ThO_2 (right). Data points directly above an asterisk correspond to diffraction peaks with an oxygen-contribution. Consistent offset of oxygen-correlated data points from the trend lines suggest that the microstrain is heterogeneous with an augmented contribution to the microstrain arising from the anion sublattice.

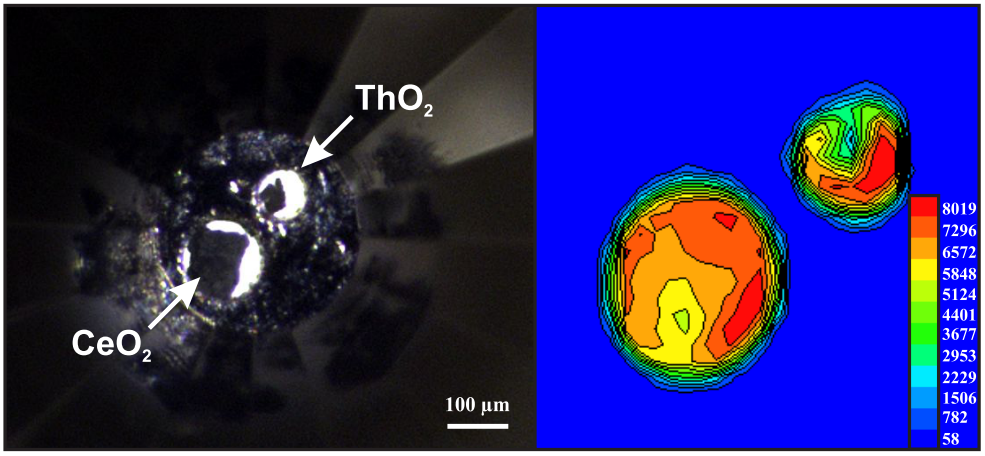
ThO2
CeO2
100 µm
8019
7296
6572
5848
5124
4401
3677
2953
2229
1506
782
58

Figure 1

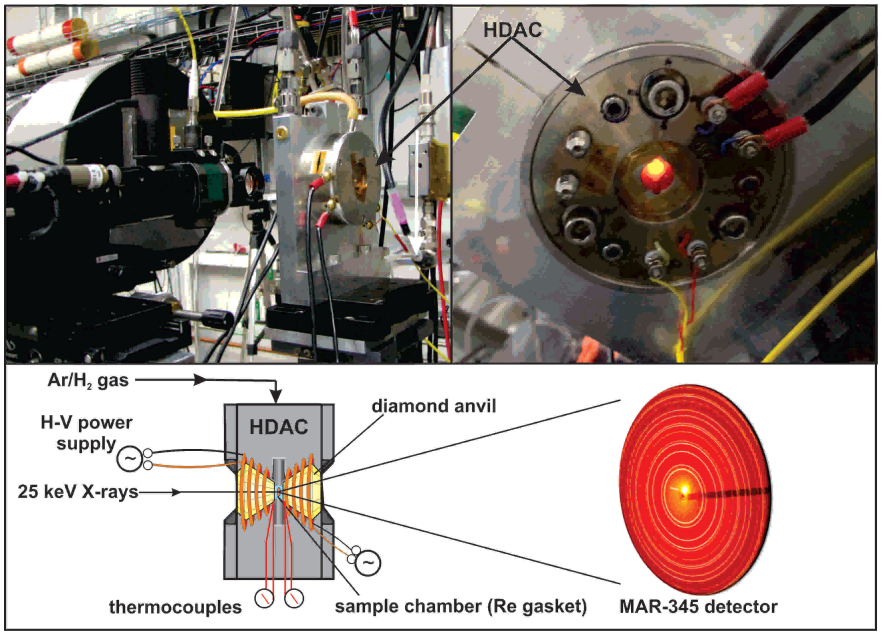
HDAC
Ar/H$_2$ gas
H-V power supply
HDAC
diamond anvil
25 keV X-rays
thermocouples
sample chamber (Re gasket)
MAR-345 detector

Figure 2

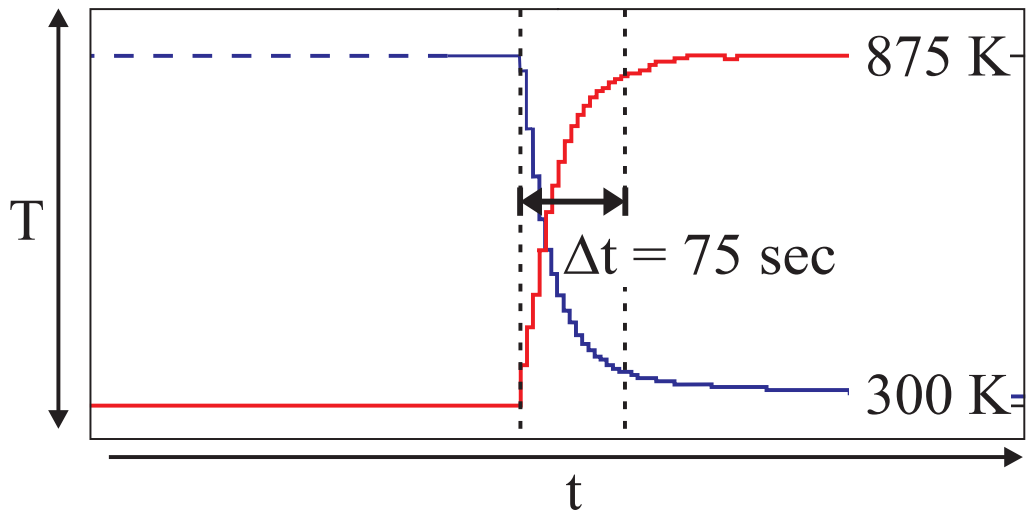

Figure 3

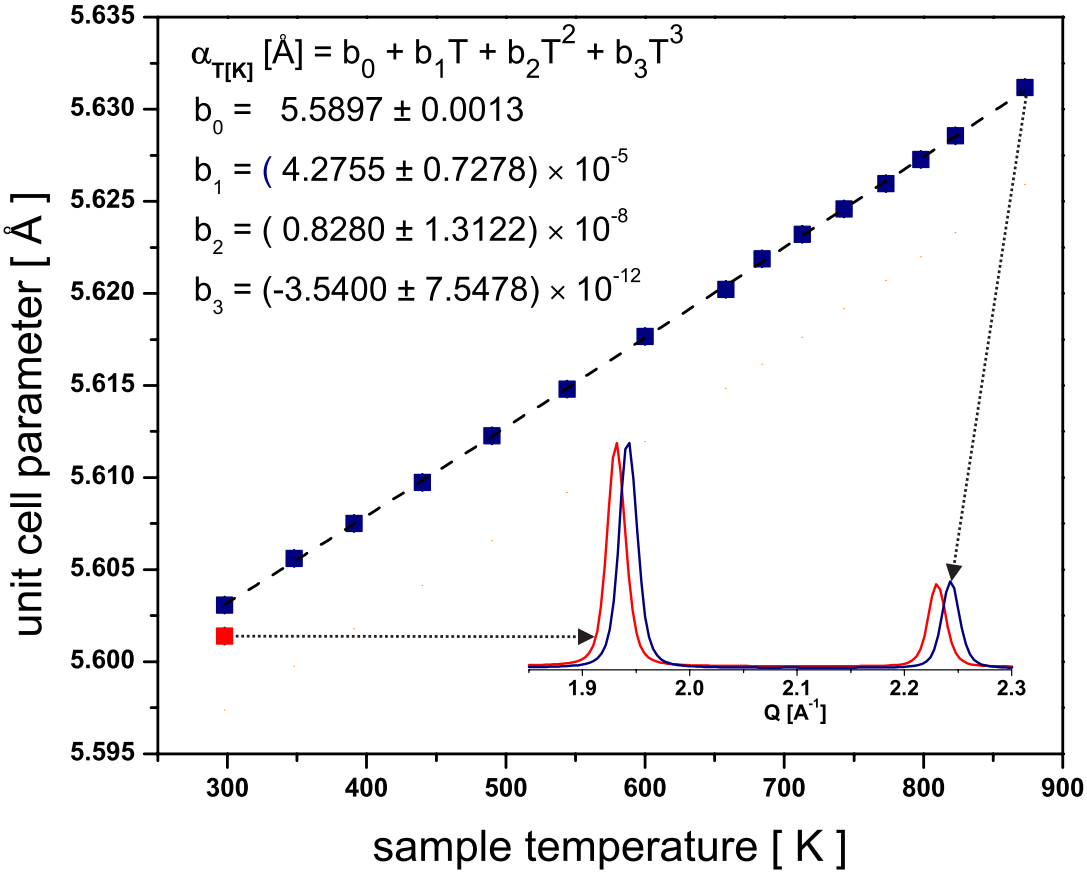
$\alpha_{T[K]}$ [Å] = $b_0 + b_1T + b_2T^2 + b_3T^3$
b_0 = 5.5897 ± 0.0013
b_1 = (4.2755 ± 0.7278) × 10^{-5}
b_2 = (0.8280 ± 1.3122) × 10^{-8}
b_3 = (-3.5400 ± 7.5478) × 10^{-12}
unit cell parameter [Å]
sample temperature [K]
Q [A^{-1}]

Figure 4

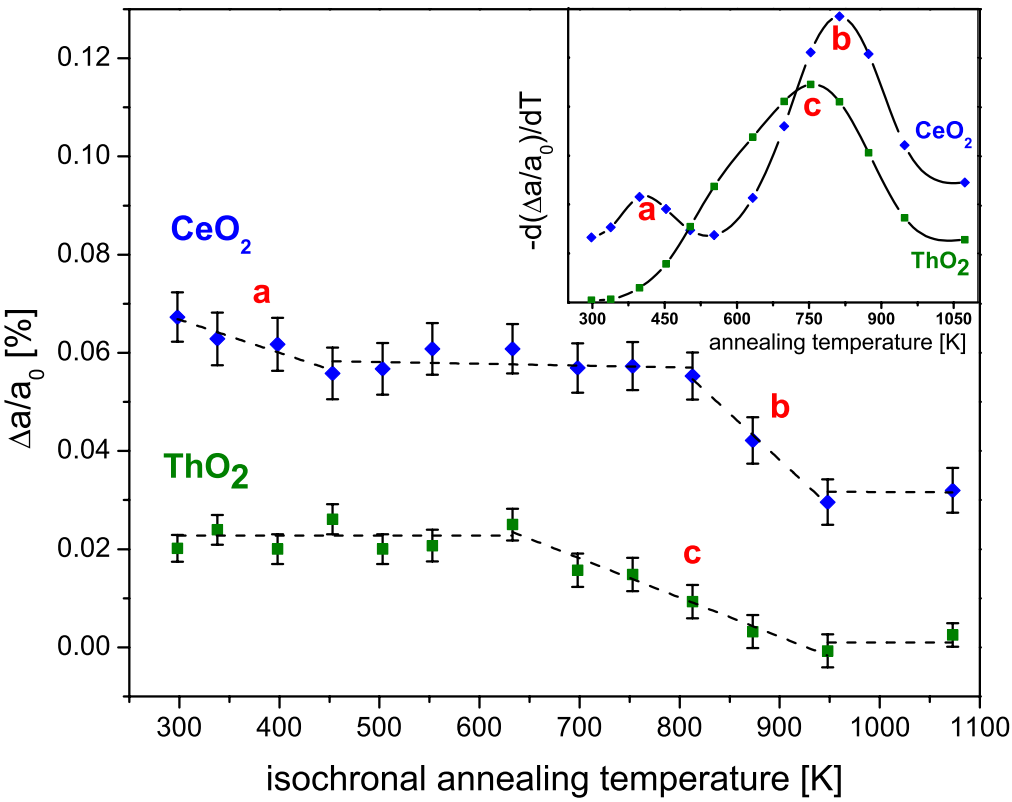
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Δa/a0 [%]
CeO2
ThO2
a
b
c
300
400
500
600
700
800
900
1000
1100
isochronal annealing temperature [K]
-d(Δa/a0)/dT
300
450
600
750
900
1050
annealing temperature [K]

Figure 5

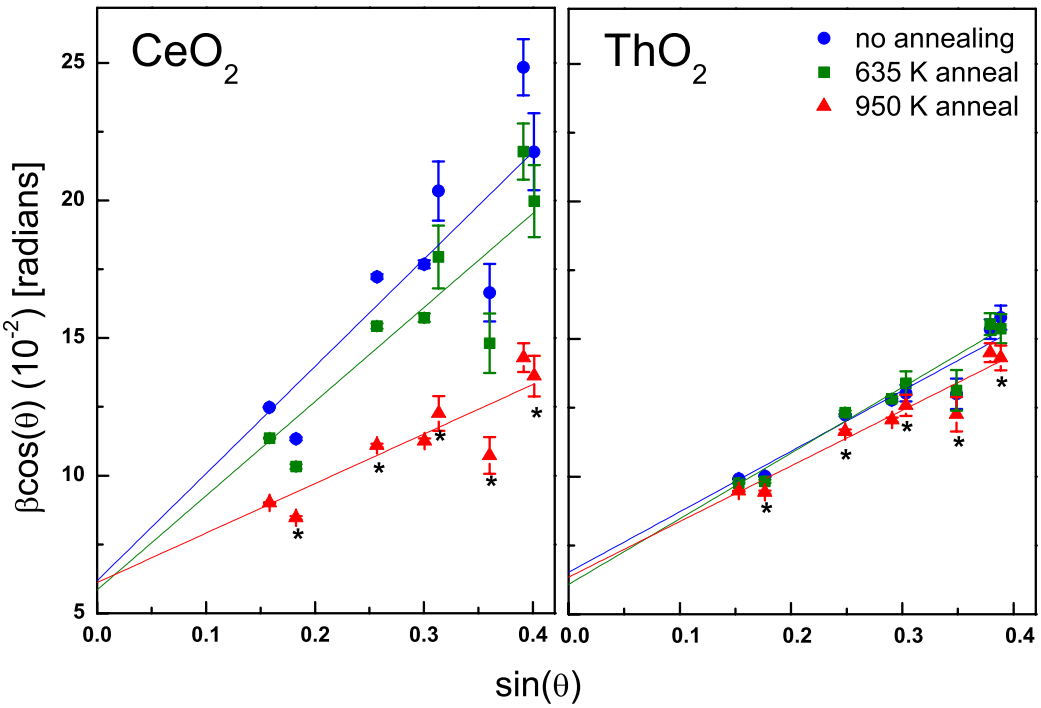
CeO2
ThO2
no annealing
635 K anneal
950 K anneal
βcos(θ) (10-2) [radians]
sin(θ)
25
20
15
10
5
0.0
0.1
0.2
0.3
0.4

Figure 6